Newman-Penrose Formalism Calculations

Stephani et al., Equation (16.49)

Metric

$$ds^2 = -e^{(-2\,g)}(\frac{\partial\,\lambda}{\partial t})^2 \mathrm{d}t \otimes \mathrm{d}t + e^{(2\,\lambda)}\mathrm{d}r \otimes \mathrm{d}r + (r^6 B^2 + 2\,r^4 AB + r^2 A^2)\mathrm{d}\theta \otimes \mathrm{d}\theta + (r^6 B^2 + 2\,r^4 AB + r^2 A^2)\sin(\theta)^2 \mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Metric Functions

$$A(t) = A$$

$$B(t) = B$$

$$g(t) = g$$

$$\lambda(r, t) = \lambda$$

Null Tetrad

Covariant components

$$l = (\sqrt{\frac{1}{2}}e^{(-g)}\frac{\partial}{\partial t}\lambda)\,dt + (-\sqrt{\frac{1}{2}}e^{(\lambda)})\,dr$$

$$n = (\sqrt{\frac{1}{2}}e^{(-g)}\frac{\partial}{\partial t}\lambda)\,dt + (\sqrt{\frac{1}{2}}e^{(\lambda)})\,dr$$

$$m = (\sqrt{\frac{1}{2}} r^3 B + \sqrt{\frac{1}{2}} r A)\, d\theta + (i \sqrt{\frac{1}{2}} r^3 B \sin(\theta) + i \sqrt{\frac{1}{2}} r A \sin(\theta))\, d\phi$$

$$\overline{m} = (\sqrt{\frac{1}{2}} r^3 B + \sqrt{\frac{1}{2}} r A)\, d\theta + (-i \sqrt{\frac{1}{2}} r^3 B \sin(\theta) - i \sqrt{\frac{1}{2}} r A \sin(\theta))\, d\phi$$

Contravariant components

$$l = (-\frac{\sqrt{2} e^g}{2 \frac{\partial}{\partial t} \lambda})\, dt + (-\frac{1}{2} \sqrt{2} e^{(-\lambda)})\, dr$$

$$n = (-\frac{\sqrt{2} e^g}{2 \frac{\partial}{\partial t} \lambda})\, dt + (\frac{1}{2} \sqrt{2} e^{(-\lambda)})\, dr$$

$$m = (\frac{1}{\sqrt{2} r^3 B + \sqrt{2} r A})\, d\theta + (\frac{i}{(\sqrt{2} r^3 B + \sqrt{2} r A) \sin(\theta)})\, d\phi$$

$$\overline{m} = (\frac{1}{\sqrt{2} r^3 B + \sqrt{2} r A})\, d\theta + (-\frac{i}{(\sqrt{2} r^3 B + \sqrt{2} r A) \sin(\theta)})\, d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2} e^g}{2 \frac{\partial}{\partial t} \lambda}\, \partial_t - \frac{1}{2} \sqrt{2} e^{(-\lambda)}\, \partial_r$$

$$\Delta = -\frac{\sqrt{2} e^g}{2 \frac{\partial}{\partial t} \lambda}\, \partial_t + \frac{1}{2} \sqrt{2} e^{(-\lambda)}\, \partial_r$$

$$\delta = \frac{1}{\sqrt{2} r^3 B + \sqrt{2} r A}\, \partial_\theta + \frac{i}{(\sqrt{2} r^3 B + \sqrt{2} r A) \sin(\theta)}\, \partial_\phi$$

$$\overline{\delta} = \frac{1}{\sqrt{2}r^3B + \sqrt{2}rA}\,\partial_\theta - \frac{i}{(\sqrt{2}r^3B + \sqrt{2}rA)\sin(\theta)}\,\partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}r^3e^{(g+\lambda)}\frac{\partial}{\partial t}B}{2\left(r^3Be^{(\lambda)}\frac{\partial}{\partial t}\lambda + rAe^{(\lambda)}\frac{\partial}{\partial t}\lambda\right)} + \frac{3\sqrt{2}r^2B\frac{\partial}{\partial t}\lambda}{2\left(r^3Be^{(\lambda)}\frac{\partial}{\partial t}\lambda + rAe^{(\lambda)}\frac{\partial}{\partial t}\lambda\right)} + \frac{\sqrt{2}re^{(g+\lambda)}\frac{\partial}{\partial t}A}{2\left(r^3Be^{(\lambda)}\frac{\partial}{\partial t}\lambda + rAe^{(\lambda)}\frac{\partial}{\partial t}\lambda\right)} + \frac{\sqrt{2}A\frac{\partial}{\partial t}\lambda}{2\left(r^3Be^{(\lambda)}\frac{\partial}{\partial t}\lambda + rAe^{(\lambda)}\frac{\partial}{\partial t}\lambda\right)}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}r^3e^{(g+\lambda)}\frac{\partial}{\partial t}B}{2\left(r^3Be^{(\lambda)}\frac{\partial}{\partial t}\lambda + rAe^{(\lambda)}\frac{\partial}{\partial t}\lambda\right)} + \frac{3\sqrt{2}r^2B\frac{\partial}{\partial t}\lambda}{2\left(r^3Be^{(\lambda)}\frac{\partial}{\partial t}\lambda + rAe^{(\lambda)}\frac{\partial}{\partial t}\lambda\right)} - \frac{\sqrt{2}re^{(g+\lambda)}\frac{\partial}{\partial t}A}{2\left(r^3Be^{(\lambda)}\frac{\partial}{\partial t}\lambda + rAe^{(\lambda)}\frac{\partial}{\partial t}\lambda\right)} + \frac{\sqrt{2}A\frac{\partial}{\partial t}\lambda}{2\left(r^3Be^{(\lambda)}\frac{\partial}{\partial t}\lambda + rAe^{(\lambda)}\frac{\partial}{\partial t}\lambda\right)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\cos(\theta)}{2\left(\sqrt{2}r^3B + \sqrt{2}rA\right)\sin(\theta)}$$

$$\beta = \frac{\cos(\theta)}{2\left(\sqrt{2}r^3B + \sqrt{2}rA\right)\sin(\theta)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{1}{4}\sqrt{2}e^{g} - \frac{\sqrt{2}e^{(-\lambda)}\frac{\partial^2}{\partial r\partial t}\lambda}{4\frac{\partial}{\partial t}\lambda}$$

$$\gamma = \frac{1}{4}\sqrt{2}e^{g} - \frac{\sqrt{2}e^{(-\lambda)}\frac{\partial^2}{\partial r\partial t}\lambda}{4\frac{\partial}{\partial t}\lambda}$$

Ricci Tensor Components

$$\begin{aligned}
\Phi_{00} = & -\frac{r^3 e^{(2\,g+2\,\lambda)}\frac{\partial}{\partial t}B\frac{\partial}{\partial t}g\frac{\partial}{\partial t}\lambda}{2\left(r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3\right)} + \frac{r^3 e^{(g+\lambda)}\frac{\partial}{\partial t}B\frac{\partial^2}{\partial r\partial t}\lambda\frac{\partial}{\partial t}\lambda}{r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3} + \frac{r^3 e^{(2\,g+2\,\lambda)}\frac{\partial}{\partial t}B\frac{\partial}{\partial t}\lambda^2}{2\left(r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3\right)} + \frac{3\,r^2 Be^{(g+\lambda)}\frac{\partial}{\partial t}\lambda^3}{r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3} \\
& + \frac{3\,r^2 B\frac{\partial}{\partial r}\lambda\frac{\partial}{\partial t}\lambda^3}{2\left(r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3\right)} - \frac{r^3 e^{(2\,g+2\,\lambda)}\frac{\partial^2}{(\partial t)^2}B\frac{\partial}{\partial t}\lambda}{2\left(r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3\right)} - \frac{3\,r^2 e^{(g+\lambda)}\frac{\partial}{\partial t}B\frac{\partial}{\partial t}\lambda^2}{r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3} + \frac{3\,r^2 B\frac{\partial^2}{\partial r\partial t}\lambda\frac{\partial}{\partial t}\lambda^2}{2\left(r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3\right)} \\
& + \frac{r^3 e^{(2\,g+2\,\lambda)}\frac{\partial}{\partial t}B\frac{\partial^2}{(\partial t)^2}\lambda}{2\left(r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3\right)} - \frac{re^{(2\,g+2\,\lambda)}\frac{\partial}{\partial t}A\frac{\partial}{\partial t}g\frac{\partial}{\partial t}\lambda}{2\left(r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3\right)} + \frac{re^{(g+\lambda)}\frac{\partial}{\partial t}A\frac{\partial^2}{\partial r\partial t}\lambda\frac{\partial}{\partial t}\lambda}{r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3} + \frac{re^{(2\,g+2\,\lambda)}\frac{\partial}{\partial t}A\frac{\partial}{\partial t}\lambda^2}{2\left(r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3\right)} \\
& - \frac{3\,rB\frac{\partial}{\partial t}\lambda^3}{r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3} + \frac{Ae^{(g+\lambda)}\frac{\partial}{\partial t}\lambda^3}{r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3} + \frac{A\frac{\partial}{\partial r}\lambda\frac{\partial}{\partial t}\lambda^3}{2\left(r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3\right)} - \frac{re^{(2\,g+2\,\lambda)}\frac{\partial^2}{(\partial t)^2}A\frac{\partial}{\partial t}\lambda}{2\left(r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3\right)} \\
& - \frac{e^{(g+\lambda)}\frac{\partial}{\partial t}A\frac{\partial}{\partial t}\lambda^2}{r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3} + \frac{A\frac{\partial^2}{\partial r\partial t}\lambda\frac{\partial}{\partial t}\lambda^2}{2\left(r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3\right)} + \frac{re^{(2\,g+2\,\lambda)}\frac{\partial}{\partial t}A\frac{\partial^2}{(\partial t)^2}\lambda}{2\left(r^3 Be^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3 + rAe^{(2\,\lambda)}\frac{\partial}{\partial t}\lambda^3\right)}
\end{aligned}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{r^6 B^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)} - \frac{r^6 B^2 \frac{\partial}{\partial r} \lambda \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)}$$

$$- \frac{r^6 B^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} \lambda^2}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)} + \frac{r^6 B^2 \frac{\partial^3}{(\partial r)^2 \partial t} \lambda \frac{\partial}{\partial t} \lambda}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)}$$

$$+ \frac{r^6 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B^2}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)} - \frac{r^4 A B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{2\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)}$$

$$- \frac{r^4 A B \frac{\partial}{\partial r} \lambda \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda}{2\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)} - \frac{r^4 A B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} \lambda^2}{2\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)}$$

$$+ \frac{r^4 A B \frac{\partial^3}{(\partial r)^2 \partial t} \lambda \frac{\partial}{\partial t} \lambda}{2\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)} - \frac{9\,r^4 B^2 \frac{\partial}{\partial t} \lambda^2}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)}$$

$$+ \frac{r^4 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} B}{2\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)} - \frac{r^2 A^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)}$$

$$- \frac{r^2 A^2 \frac{\partial}{\partial r} \lambda \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)} - \frac{r^2 A^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} \lambda^2}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)}$$

$$+ \frac{r^2 A^2 \frac{\partial^3}{(\partial r)^2 \partial t} \lambda \frac{\partial}{\partial t} \lambda}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)} - \frac{3\,r^2 A B \frac{\partial}{\partial t} \lambda^2}{2\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)}$$

$$+ \frac{r^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A^2}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)} - \frac{A^2 \frac{\partial}{\partial t} \lambda^2}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)}$$

$$+ \frac{e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^2)}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$
\begin{aligned}
\Phi_{22} = & -\frac{r^3 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{2\,(r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^3 e^{(g+\lambda)} \frac{\partial}{\partial t} B \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda}{r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3} + \frac{r^3 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial}{\partial t} \lambda^2}{2\,(r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{3\,r^2 B e^{(g+\lambda)} \frac{\partial}{\partial t} \lambda^3}{r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3} \\
& + \frac{3\,r^2 B \frac{\partial}{\partial r} \lambda \frac{\partial}{\partial t} \lambda^3}{2\,(r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^3 e^{(2\,g+2\,\lambda)} \frac{\partial^2}{(\partial t)^2} B \frac{\partial}{\partial t} \lambda}{2\,(r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{3\,r^2 e^{(g+\lambda)} \frac{\partial}{\partial t} B \frac{\partial}{\partial t} \lambda^2}{r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3} + \frac{3\,r^2 B \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{2\,(r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& + \frac{r^3 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial^2}{(\partial t)^2} \lambda}{2\,(r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{2\,(r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r e^{(g+\lambda)} \frac{\partial}{\partial t} A \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda}{r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3} + \frac{r e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} \lambda^2}{2\,(r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& - \frac{3\,r B \frac{\partial}{\partial t} \lambda^3}{r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3} - \frac{A e^{(g+\lambda)} \frac{\partial}{\partial t} \lambda^3}{r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3} + \frac{A \frac{\partial}{\partial r} \lambda \frac{\partial}{\partial t} \lambda^3}{2\,(r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r e^{(2\,g+2\,\lambda)} \frac{\partial^2}{(\partial t)^2} A \frac{\partial}{\partial t} \lambda}{2\,(r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& + \frac{e^{(g+\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} \lambda^2}{r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3} + \frac{A \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{2\,(r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial^2}{(\partial t)^2} \lambda}{2\,(r^3 B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r A e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = {} & \frac{r^6 B^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda^2}{12\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^6 B^2 \frac{\partial}{\partial r} \lambda \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{12\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^6 B^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} \lambda^3}{12\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& + \frac{r^6 B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^6 B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^6 B^2 \frac{\partial^3}{(\partial r)^2 \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{12\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& + \frac{r^5 B^2 \frac{\partial}{\partial r} \lambda \frac{\partial}{\partial t} \lambda^3}{2\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^6 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B^2 \frac{\partial}{\partial t} \lambda}{12\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^6 B e^{(2\,g+2\,\lambda)} \frac{\partial^2}{(\partial t)^2} B \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& + \frac{r^4 A B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^5 B^2 \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{2\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^4 A B \frac{\partial}{\partial r} \lambda \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& + \frac{r^4 A B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} \lambda^3}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^6 B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial^2}{(\partial t)^2} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^4 B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& + \frac{r^4 A e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^4 B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^4 A e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& - \frac{r^4 A B \frac{\partial^3}{(\partial r)^2 \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{7\,r^4 B^2 \frac{\partial}{\partial t} \lambda^3}{4\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{2\,r^3 A B \frac{\partial}{\partial r} \lambda \frac{\partial}{\partial t} \lambda^3}{3\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& + \frac{r^4 B e^{(2\,g+2\,\lambda)} \frac{\partial^2}{(\partial t)^2} A \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^4 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} B \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^4 A e^{(2\,g+2\,\lambda)} \frac{\partial^2}{(\partial t)^2} B \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& + \frac{r^2 A^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda^2}{12\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{2\,r^3 A B \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{3\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^2 A^2 \frac{\partial}{\partial r} \lambda \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{12\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& + \frac{r^2 A^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} \lambda^3}{12\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^4 B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial^2}{(\partial t)^2} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^4 A e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial^2}{(\partial t)^2} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& + \frac{r^2 A e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^2 A e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^2 A^2 \frac{\partial^3}{(\partial r)^2 \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{12\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& - \frac{3\,r^2 A B \frac{\partial}{\partial t} \lambda^3}{2\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r A^2 \frac{\partial}{\partial r} \lambda \frac{\partial}{\partial t} \lambda^3}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A^2 \frac{\partial}{\partial t} \lambda}{12\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& + \frac{r^2 A e^{(2\,g+2\,\lambda)} \frac{\partial^2}{(\partial t)^2} A \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r A^2 \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^2 A e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial^2}{(\partial t)^2} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} \\
& - \frac{A^2 \frac{\partial}{\partial t} \lambda^3}{12\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3}{12\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}
\end{aligned}
$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$\Psi_2 = -\frac{r^6 B^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^6 B^2 \frac{\partial}{\partial r} \lambda \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^6 B^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} \lambda^3}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$+ \frac{r^6 B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^6 B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$+ \frac{r^6 B^2 \frac{\partial^3}{(\partial r)^2 \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^5 B^2 \frac{\partial}{\partial r} \lambda \frac{\partial}{\partial t} \lambda^3}{2\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^6 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B^2 \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$+ \frac{r^6 B e^{(2\,g+2\,\lambda)} \frac{\partial^2}{(\partial t)^2} B \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^4 A B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda^2}{3\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^5 B^2 \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{2\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$- \frac{r^4 A B \frac{\partial}{\partial r} \lambda \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{3\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^4 A B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} \lambda^3}{3\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^6 B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial^2}{(\partial t)^2} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$+ \frac{r^4 B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^4 A e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^4 B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$+ \frac{r^4 A e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^4 A B \frac{\partial^3}{(\partial r)^2 \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{3\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^4 B^2 \frac{\partial}{\partial t} \lambda^3}{2\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$+ \frac{2\,r^3 A B \frac{\partial}{\partial r} \lambda \frac{\partial}{\partial t} \lambda^3}{3\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^4 B e^{(2\,g+2\,\lambda)} \frac{\partial^2}{(\partial t)^2} A \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^4 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} B \frac{\partial}{\partial t} \lambda}{3\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$+ \frac{r^4 A e^{(2\,g+2\,\lambda)} \frac{\partial^2}{(\partial t)^2} B \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^2 A^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{2\,r^3 A B \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{3\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$- \frac{r^2 A^2 \frac{\partial}{\partial r} \lambda \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^2 A^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} \lambda^3}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^4 B e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial^2}{(\partial t)^2} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$- \frac{r^4 A e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} B \frac{\partial^2}{(\partial t)^2} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^2 A e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} g \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r^2 A e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$+ \frac{r^2 A^2 \frac{\partial^3}{(\partial r)^2 \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} + \frac{r A^2 \frac{\partial}{\partial r} \lambda \frac{\partial}{\partial t} \lambda^3}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^2 e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A^2 \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$+ \frac{r^2 A e^{(2\,g+2\,\lambda)} \frac{\partial^2}{(\partial t)^2} A \frac{\partial}{\partial t} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r A^2 \frac{\partial^2}{\partial r \partial t} \lambda \frac{\partial}{\partial t} \lambda^2}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{r^2 A e^{(2\,g+2\,\lambda)} \frac{\partial}{\partial t} A \frac{\partial^2}{(\partial t)^2} \lambda}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$+ \frac{A^2 \frac{\partial}{\partial t} \lambda^3}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)} - \frac{e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3}{6\,(r^6 B^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + 2\,r^4 A B e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3 + r^2 A^2 e^{(2\,\lambda)} \frac{\partial}{\partial t} \lambda^3)}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"